INDIA AI IMPACT FESTIVAL 2026 · AI IMPACT CREATORS (STUDENT) · SYNOPSIS

SAILAAB · Punjab Flood Intelligence

Open SAR flood mapping · decade hazard atlas · impact analytics · district flood forecasting · live monitoring

ਸੈਲਾਬ · ਪੰਜਾਬ ਦੀ ਹੜ੍ਹ ਖੁਫ਼ੀਆ ਪ੍ਰਣਾਲੀ

LIVE: bakathefish.github.io/Flood CODE + DATA: github.com/bakathefish/Flood

11	**467**	**105,183**	**91**	**ρ=0.72**	**0**	**0**
MONSOONS MAPPED	FLOOD DAYS PROCESSED	HA FLOODED 2025 (TIER-A)	TEHSILS SCORED	VS GOVT GIRDAWARI	CWC STATIONS IN PUNJAB	LOGINS REQUIRED

01 The problem

In August and September 2025, Punjab suffered its worst flood since 1988: all 23 districts affected, ~3.55 lakh people, 1.48–1.75 lakh hectares of crops destroyed. Four failures made it worse than it had to be. Nobody was forecasting: in the Central Water Commission's own state-wise table, Punjab has zero flood-forecasting stations, absent from a 226-station national network (Haryana 1, J&K 3; the one site CWC ever added on the Ravi is defunct). Punjab's rivers sit outside India's flood-forecast network. Maps existed but were locked: ISRO's NDEM produced excellent flood maps as static PDFs, one snapshot at a time, not analyzable, not open. Nobody knew the recurrence: Punjab has no public flood-frequency atlas (NRSC published them for Assam and Bihar), so villages that flooded four times in a decade were treated as first-time surprises. Relief ran on foot: girdawari compensation required weeks of manual crop surveys while satellite data that could target it sat unused.

02 What Sailaab is

A five-module open pipeline that turns free satellite radar into flood intelligence a district officer can act on. I built it in Punjab during the 2026 monsoon, and it is running live today (the monitor executes on public CI every 6 hours with zero accounts or keys; jurors can watch it commit).

MODULE	WHAT IT PRODUCES
2025 Flood Atlas	Sentinel-1 SAR flood extent (physics baseline + Random Forest), per-district and per-tehsil statistics
Decade Hazard Atlas	Every monsoon 2015–2025 mapped identically → Punjab's first public flood-frequency map + named "repeat victims" tehsil list
Impact Engine	Crop damage (₹351–523 crore band), population exposure, submergence-duration atlas, the rain×dams×releases causal chart
Forecaster	XGBoost district flood-risk model trained on the pipeline's own decade of SAR-derived labels, SHAP-explained and conformal-bounded
Live Monitor	Every new Sentinel-1 pass → district flood km² → ਪੰਜਾਬੀ / हिन्दी / English alerts, automatically

91 tehsils are individually scored, giving relief a named list: Khadur Sahib (Tarn Taran) flooded in 6 of the last 11 monsoons, Sultanpur Lodhi (Kapurthala) in 5, Moonak (Sangrur) and Patti (Tarn Taran) in 4. These are the exact administrative units where girdawari verification and pre-positioning should start. Population exposure adds the human denominator: 0.76–1.78 lakh people lived inside the mapped water (GHSL 2025), consistent with the official 3.55 lakh "affected" once you see that flooded land is overwhelmingly low-density cropland (146–206 people/km² vs Punjab's ~550 mean). Everything is login-free and reproducible: anonymous Planetary Computer STAC for Sentinel-1, keyless Copernicus-GFM WMS, IMD rainfall, CWC reservoir records. A district collector, a journalist, or a student can re-run every number with zero accounts.

03 The AI, and why it is honest

Random Forest flood classifier. Features: VV/VH backscatter, pre/post change, terrain. Labels: bootstrapped from *agreement* between our physics-threshold map and Copernicus GFM. No hand digitization, and we say so. Validation is spatial cross-validation across river basins (train Ravi–Beas districts → test Sutlej districts, and swap). We report the trap honestly: agreement-strata points are near-separable (fold F1 ≈ 1.0), so we publish the independent fresh-point comparison (OA 0.983 vs GFM) and the three-method envelope (Tier-A 34k < RF 52k < GFM 86k ha in-district). A small U-Net trained on the Sen1Floods11 hand-labeled benchmark (test IoU 0.63) completes a three-method benchmark table in which the target-domain RF wins and the imported deep model's resolution gap is quantified rather than hidden.

The self-labeling loop. Nobody hands you 11 years of Punjab flood labels, so the mapping engine manufactured its own: 1,177 day-slots probed, 467 flood-active days rasterized, 2,420 district-window training rows. Doing that exposed a trap that would silently poison any naive attempt: June and July "floods" in Punjab are largely rice transplanting, because deliberately flooded paddies inflate apparent flood area ~20×. All forecaster training excludes the transplant windows; calibrated late-season products ship alongside raw ones.

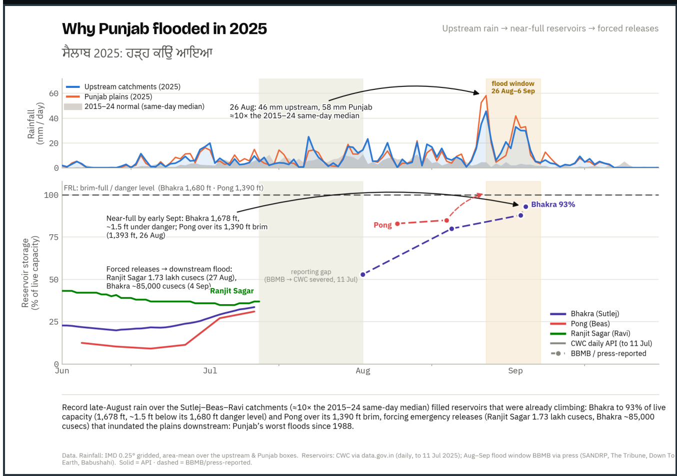
Daily district forecaster (rebuilt after submission; see the note below). Unit: district × day, every day of the monsoon, 2015–2025: 23,540 district-days built from 478 cached Copernicus GFM passes. The question is operational: on this day, will the satellite see flooding above 0.5% of this district's area within the next three days? Predictors are satellite flood history only, ten features: two district priors, observed extent now and over three days, position in the season, neighbouring districts, a district-week climatology, and three self-excitation terms carrying recent flooding across the district adjacency graph with ~3-day exponential decay, matched to routing time. Gradient boosting (200 trees, depth 3) rather than deep learning: 272 positive district-days will not train an LSTM. *Rainfall was tested as a feature family and measurably did not help, so it was dropped*, and the live system therefore needs no rain feed at all.

> The result, and what it is not. Validation is walk-forward: every season is forecast using only the seasons before it, and the alert level is set inside each fold rather than from scores the model has already fitted. Over 2019–2025 (8,863 district-days, 272 positives, base rate 3.07%) it reaches pooled average precision 0.249, 8.1× the base rate, ahead of every baseline tested: persistence 0.037, a transparent climatology-plus-neighbour rule 0.040, penalised logistic 0.075, Firth 0.092, and the same model without the excitation features 0.138. That pooled figure is not its typical season: averaged season by season the mean is 0.176 and the median 0.083. It wins clearly in 2025 (0.536 vs 0.100 for persistence); in 2023 the no-excitation variant beats it, 0.243 to 0.145. At the deployed operating point it raises about 24 alerts a season, roughly one in three followed by flooding within three days, catching about one recorded onset in four. This is a selective trigger for satellite-detected flooding, not a comprehensive warning system. All of it is retrospective and post-selection; the 2026 monsoon, running now under a frozen protocol, is the prospective test.

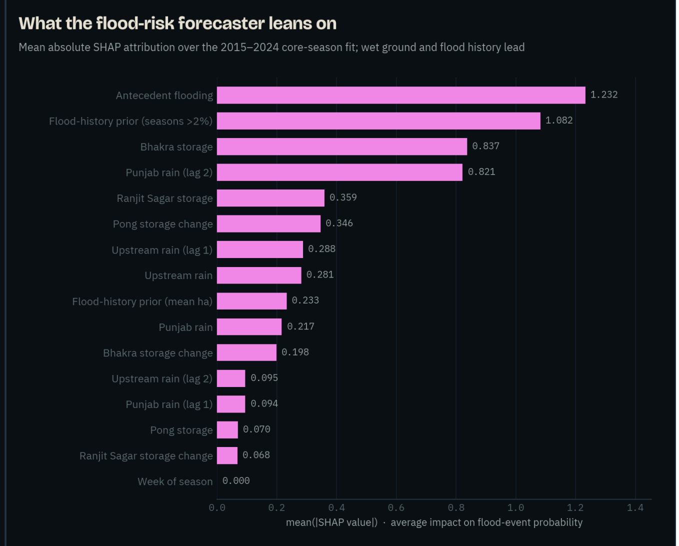
The audits we ran on ourselves, including the ones that cost us headlines. The dam ablation was pre-declared and both expectations failed, shipped verbatim: SHAP ranked Bhakra storage #3 in attribution, yet deleting all six reservoir features left the 2025 hindcast unchanged, so the dam signature was attribution rather than load-bearing skill, and the dam story lives in the headroom analysis where the evidence supports it. Rebuilding at daily resolution cost two more. An early claim of *96% alert precision at four alerts a season was withdrawn*: the threshold had been set on the model's own training scores, which it had already fitted, and recomputed out-of-fold it is about a third. A claim that the model *beat every baseline in every flood season was withdrawn* as simply false. A label bug that scored partially-observed horizons as negatives, crediting the model for floods the satellite never looked for, was found and censored. Every one of those corrections is in the repository rather than quietly dropped.

Vs Google Flood Hub: river-stage forecasts from an LSTM trained across thousands of gauges; it under-models dam-*regulated* rivers and outputs no district crop risk. That architecture is not available at this sample size, and we say so rather than dismissing it: with 96 recorded onsets, the self-exciting formulation is what the data supports. Sailaab is impact-native and regulation-aware in its analytics. Complementary, not competing.

Note on this document (updated 6 Aug 2026). The forecaster above was rebuilt after the submitted synopsis was written. The submitted text describes the earlier model: district × 10-day window, rainfall and reservoir predictors, leave-one-year-out validation. It was superseded for three documented reasons: leave-one-year-out lets a model train on seasons that had not happened yet at issue time; 10-day windows discarded daily satellite masks already on disk; and its rainfall and reservoir features did not carry the skill attributed to them. The submitted document stands as submitted; this repository carries the system actually running, with the retractions published alongside it.



Why it flooded: ~10× rain spike (26 Aug) over near-full reservoirs → forced releases (Ranjit Sagar 1.73 lakh cusecs, 27 Aug) → inundation. IMD grids + CWC/BBMB records, including the reporting gap.



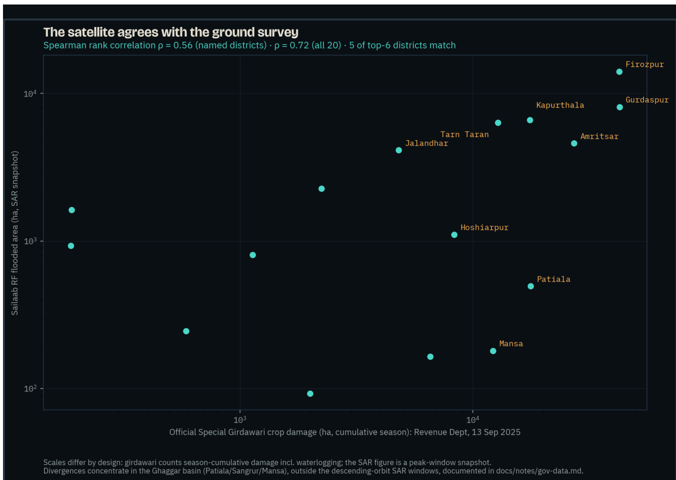
Why the model says what it says: SHAP attribution: wet ground and flood history dominate; Bhakra storage ranks #3 (ablation-tested above: attribution, not load-bearing skill).

04 Validated three independent ways, failures kept

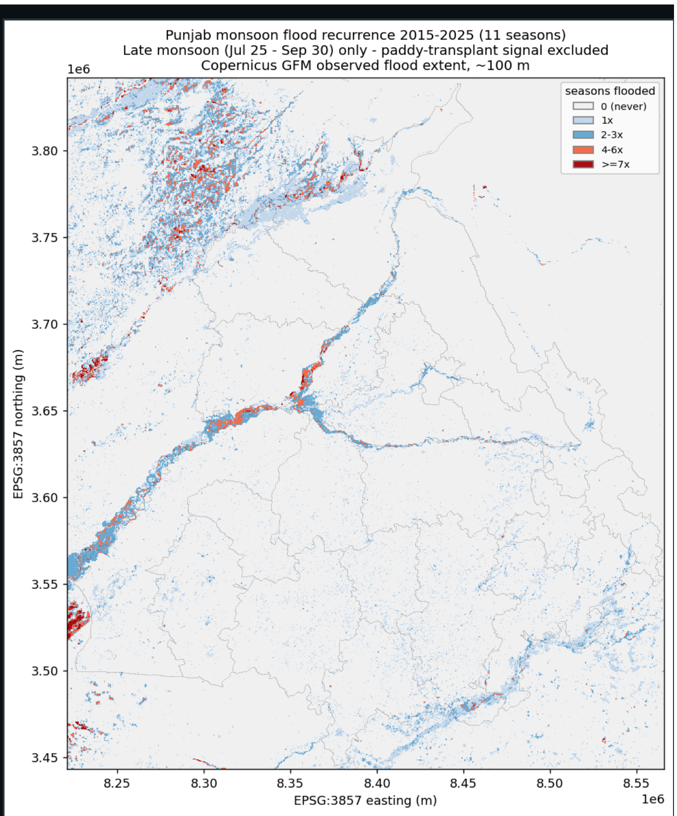
Every satellite step is gated by a pre-declared checkpoint: the expected numeric band is written into the verification log *before* the run, then the actual, then PASS/FAIL. The log ships in the repo with its failures intact (e.g., the NRSC Sangrur-2023 anchor fails on its exact date because no satellite pass existed that day; the same event registers at 33,771 ha in adjacent windows). Validation is triangulated:

- Copernicus GFM (quantitative): fresh-point confusion matrices; RF sits inside its Tier-A/GFM parent envelope on both precision and recall.
- Sentinel-2 optical truth points (independent physics that breaks SAR-chain circularity): 237 photo-interpreted NDWI points confirm 81.7% of RF flood pixels as standing water two weeks post-peak (pre-declared gate ≥60%); all pre-declared precision/recall orderings confirmed.
- ISRO NDEM sheets (visual): approximate-extent cross-check, stated as such; no pixel agreement claimed.

> The ground survey agrees. Against the Revenue Department's Special Girdawari district table (13 Sep 2025), our satellite ranking scores Spearman ρ = 0.72 across all 20 districts (0.56 over the 16 named), with 5 of the top-6 districts identical; divergences concentrate in the Ghaggar basin, outside our SAR windows, and are documented. ₹ value-at-risk was rebuilt on the government's own DES district paddy yields (₹523 crore, within about 4% of the independent estimate); the per-pixel submergence-duration atlas (51,202 ha under water for 7 days or more) refines damage to a duration-weighted ₹351 crore. Independent context: IMD's own grids put Aug 20–Sep 5 2025 rainfall at +9.7σ (rank 1 of 11 years), and a pre-registered 65-year Mann-Kendall analysis shows the loading largely stationary except a significantly rising frequency of extreme-wet days upstream (p=0.017); 2025 was a compound timing event, not merely a rainfall record. A pre-declared headroom analysis quantifies the management component honestly: on the eve of the releases, Pong and Ranjit Sagar held +1.0 and +0.45 BCM more than their own decade-median filling practice (Bhakra was near-median), about 1.68 BCM less pre-positioned buffer, worth one to four days of peak release. Real, but second-order to the rain.



Satellite vs ground survey: district ranks vs the Special Girdawari table.



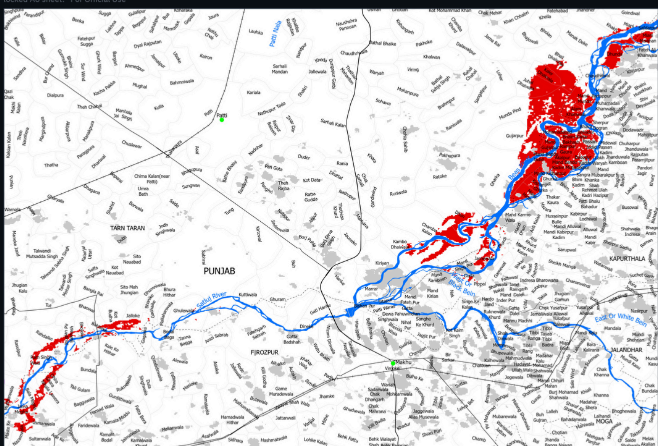
Punjab's first public flood-frequency atlas (2015–2025, transplant-calibrated).

The same flood, two access models

SAILAAB
Punjab flood intelligence

Punjab, August 2025: the Beas breaks its banks across the Kapurthala doab. ISRO mapped it; so did we.

ISRO NDEM · 19 Aug 2025 · static PDF
locked A0 sheet: "For Official Use"



Map excerpt: NDEM / NRSC / ISRO, ndem.nrsc.gov.in, Govt. of India map product, downscaled for comparison/criticism with credit.

SAILAAB · same flood · open, interactive, reproducible
Sentinel-1 SAR → random-forest mask · EPSG:32643 · MIT + open data

SAILAAB RF flood mask (rf_flood_2025.tif) over Punjab district polygons (datameet, ODbL). Cyan = inundation.

Same event, two access models. NDEM sheets validated our extent visually; full georeferenced comparison on the roadmap.

Approximate extent match: the two halves are aligned by district shape, not georeferenced (the NDEM sheet is a projection-less raster PDF). Left: Resourcesat-2A AWiFS, 19-08-2025, Kapurthala & Tarn Taran (MAP ID 2025/FL/PB/2/19082025).

Right: SAILAAB flood-2025 RF mask, Beas–Sutlej doab window lon[74.40, 75.64] × lat[30.78, 31.50].

Same flood, two access models: ISRO NDEM's locked PDF sheet (left) vs Sailaab's open, interactive, reproducible map (right). NDEM sheets validated our extent visually; full georeferencing is on the roadmap. Map excerpt: NDEM/NRSC/ISRO.

05 Running right now

The live monitor is not a plan. It executes on GitHub's public runners every 6 hours with zero secrets: it detects each new Sentinel-1 pass over Punjab, computes district flood areas, renders trilingual alerts, and commits the state publicly. The public app is interactive: an every-district map with five switchable layers (live observed, live model risk, 2025 flood, decade frequency, hindcast risk), click-through district panels (flooded ha, crop loss, ₹ at risk, recurrence, hindcast P, official girdawari figure), a before/after satellite swipe, a "replay 2025" scrubber that shows model risk rising before the water arrives, and a full ਪੰਜਾਬੀ/हिन्दी interface toggle. From 25 July 2026, inside judging week, the forecaster also publishes live daily district ranking scores on every CI cycle (it activates only in its trained post-transplant domain; the countdown is shown honestly until then).

06 Impact, users, SDGs

- SDMA / district EOCs: recurrence zones + live district km^2 for pre-positioning and evacuation priority.
- Revenue Dept / girdawari: flood-extent $\cap$ cropland per tehsil turns weeks of manual survey into verification; 20 print-ready district flood briefs ship in the repo.
- Insurers (PMFBY), banks, farmers: independent reproducible loss extents; vernacular alerts; every map open.
- Worked example (one click in the app): Sultanpur Lodhi tehsil, Kapurthala. Its district was flagged #1 by the model in the Aug 14–24 window; 4,733 ha flooded (4,028 ha cropland); it has flooded in 5 of the last 11 monsoons. Its girdawari list and a ਪੰਜਾਬੀ alert are one click away: the relief workflow, demonstrated on real 2025 data. Outreach to PSDMA and the worst-hit District Collectors is drafted for dispatch with this submission.
- SDGs: 2 (crop-loss quantification), 11 (resilient communities), 13 (climate adaptation).

07 Originality

(1) First open ML-ready flood mapping of Indian-Punjab 2025 (the published RF precedent covered the Pakistani side); (2) Punjab's first public flood-frequency atlas and first submergence-duration atlas; (3) a self-labeling SAR→ML loop that manufactures its own decade of training data, with the rice-transplant contamination discovery published; (4) a daily district forecaster validated walk-forward at 8.1× the base rate, with the headline numbers it failed to survive published as retractions; (5) an end-to-end account-free architecture, so every number re-runs without an account; (6) the quantified forecast gap: Punjab is absent from CWC's own station network, and Sailaab is the district-level layer that gap leaves open. The negative results (a challenger model that lost; a U-Net losing to the target-domain RF; the ablation's two failed expectations; a retracted 96% precision headline; rainfall failing to help at all) are published with the same prominence as the wins.

08 Roadmap & openness

Multi-state scale-out (the pipeline is a bounding box + district file), village-circle girdawari integration, Bhashini voice alerts, GPU-scale models via the IndiaAI Compute programme, EOS-04 Indian-SAR cross-validation via NRSC Bhoonidhi, and official partnerships. Code MIT; maps and tables CC-BY-4.0; 450+ automated tests; the method paper, data-source registry (every dataset: URL, license, access date) and the full verification log live in the repository.

11 monsoons · 467 flood days · 105,183 ha mapped · 91 tehsils scored · ρ=0.72 vs the government girdawari · 3 languages · 0 CWC forecast stations in Punjab · 0 logins required · github.com/bakathefish/Flood · bakathefish.github.io/Flood · Contains modified Copernicus Sentinel & CEMS-GFM data